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1. Consider the nonlinear equation $y = f(x) = x^2 + 2$, where the input x is a Gaussian random variable having mean of 1 and variance of 1. We desire to calculate the statistics of the output variable y using the sigma-point approach, where we use the CDKF parameter values when doing so, and assume that $h = \sqrt{3}$. 1 / 1 point

What is the value of $\alpha_0^{(m)}$? Enter your answer with two digits to the right of the decimal point.

0.67

✔ Correct
Yes. That is the right answer.

2. For the same problem described in question 1, what is the value you compute for $\bar{y}$? Enter your answer with one digit to the right of the decimal point, but use full precision for all intermediate values. 1 / 1 point

4

✔ Correct
Yes. That is the correct answer. Note that this turns out to be the exact analytic result in this case as well!

3. For the same problem described in question 1, what is the value you compute for $\Sigma_{\tilde{y}}$? Enter your answer with one digit to the right of the decimal point. 1 / 1 point

6

✔ Correct
Yes. That is the correct answer. Note that it also turns out to be the exact analytic result in this case!